**Artificial Intelligence & Machine Learning – Task 3**

**Model Validation, Overfitting Control & Hyperparameter Tuning**

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**Organization:** Maincrafts Technology
**Task:** Model Validation, Overfitting Control & Hyperparameter Tuning
**Dataset:** California Housing Dataset

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## 1. Introduction

In Task-2, multiple regression models were trained and compared.
Task-3 moves deeper into professional Machine Learning practices by focusing on:

- Detecting overfitting
- Applying cross-validation
- Performing hyperparameter tuning
- Selecting the most reliable and generalizable model

The goal is not only to improve accuracy but to ensure **model reliability and stability on unseen data**.

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## 2. Dataset Overview

The same **California Housing Dataset** was used to maintain consistency across tasks.

- Target Variable: Median House Value
- Features: Income, Rooms, Population, Occupancy, Latitude, Longitude

Using the same dataset ensures fair comparison between baseline and optimized models.

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## 3. Detecting Overfitting

To analyze overfitting, a **Decision Tree Regressor** was trained and evaluated on:

- Training Data
- Testing Data

**Observation:**

A large difference between **Training RMSE** and **Testing RMSE** indicates overfitting.

In the baseline Decision Tree model:

- Training error was very low
- Testing error was significantly higher

This confirms that the model was memorizing training data rather than generalizing well.

**Interpretation:**

Tree-based models without constraints often overfit.
Controlling model complexity is essential.

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**4. Cross-Validation for Reliable Evaluation**

Instead of relying on a single train-test split, **5-Fold Cross-Validation** was applied.

**Why Cross-Validation?**

- Reduces dependency on one data split
- Provides stable performance estimation
- Reflects real-world reliability

The model was evaluated using:

- RMSE
- $R^2$ Score

The average cross-validated RMSE gave a more realistic estimate of performance compared to single split evaluation.

This aligns with industry best practices for model validation.

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**5. Hyperparameter Tuning using GridSearchCV**

To reduce overfitting and improve performance, hyperparameters of the Decision Tree were tuned using **GridSearchCV**.

**Parameters Tuned:**

- max_depth
- min_samples_split

**Why Hyperparameter Tuning?**

- Controls model complexity
- Reduces variance
- Improves generalization

GridSearchCV systematically tested multiple parameter combinations using cross-validation and selected the best configuration.

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**6. Evaluation of Optimized Model**

After tuning:

- Training and testing RMSE gap reduced
- Cross-validation score improved
- Model stability increased

The optimized Decision Tree performed better than the untuned version and showed improved generalization.

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**7. Model Comparison**

The following models were compared:

- Linear Regression (Baseline – Task 1)
- Ridge Regression (Task 2)
- Tuned Decision Tree (Task 3)

**Comparison Criteria:**

- RMSE
- $R^2$ Score
- Cross-Validation Performance
- Stability

The tuned model showed improved reliability compared to the baseline models.

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**8. Final Model Selection Justification**

The final selected model was chosen based on:

1. Balanced RMSE and $R^2$ performance

2. Reduced overfitting gap
3. Stable cross-validation scores
4. Controlled model complexity

While Linear Regression is simple and interpretable, the tuned Decision Tree captured non-linear relationships more effectively without excessive overfitting.

The decision was based on **generalization performance**, not just training accuracy.

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**9. Key Learnings from Task-3**

This task strengthened understanding of:

- Overfitting vs Underfitting
- Bias-Variance Tradeoff
- Cross-Validation techniques
- Hyperparameter tuning strategies
- Scientific model selection

It reflects real-world Machine Learning workflows used in production systems.

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**10. Conclusion**

Task-3 successfully demonstrated professional Machine Learning practices:

- Overfitting detection
- Reliable validation
- Hyperparameter optimization
- Objective model comparison

The optimized model showed improved generalization and reliability, aligning with industry standards for robust ML systems.

This task enhanced practical experience in model validation and tuning, moving beyond basic model training.

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**11. Tools & Technologies Used**

- Python

- Pandas & NumPy
- Scikit-learn
- Matplotlib
- Jupyter Notebook